The Synapse

Exam code: 0610 & 0970

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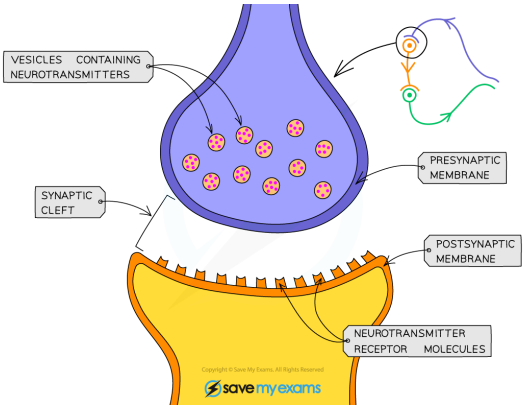
Help me understand

Structure of a Synapse: Extended

- The junction between two neurones is known as a **synapse**



A synapse

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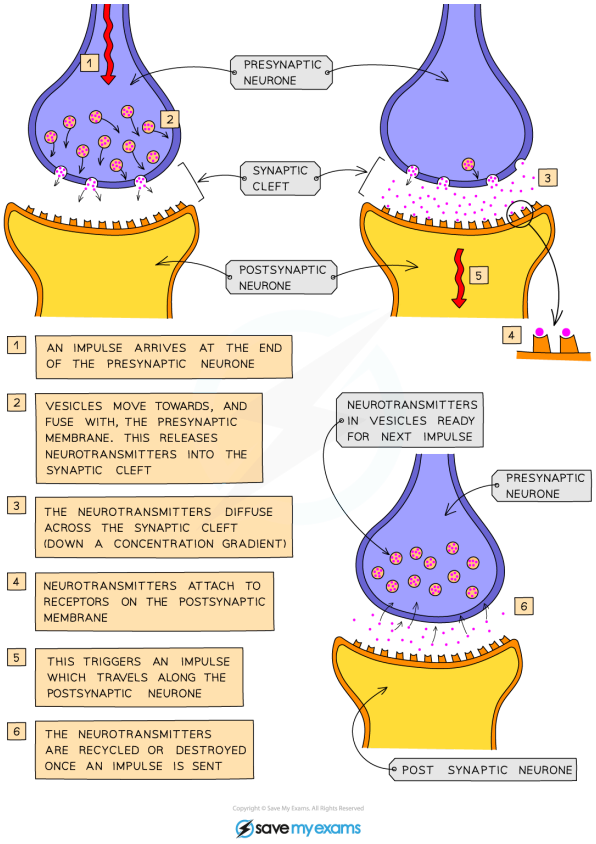
Synapses & Neurotransmitters:

- Neurones never touch each other
- The junctions (gaps) in between them are called **synapses**
- The electrical impulse travels along the first axon
- This triggers the nerve-ending of the presynaptic neurone to release **chemical** messengers called **neurotransmitters** from vesicles which fuse with the presynaptic membrane
- The neurotransmitters **diffuse** across the synaptic gap (or cleft) and **bind with receptor molecules** on the membrane of the second neurone (known as the **postsynaptic membrane**)
- This **stimulates** the second neurone to generate an electrical impulse that travels down the second axon
- The neurotransmitters are then **destroyed to prevent continued stimulation** of the second neurone which would cause repeated impulses to be sent
- Synapses ensure that impulses **only travel in one direction**, avoiding confusion within the nervous system if impulses were travelling in both directions
- As this is the only part of the nervous system where messages are **chemical** as opposed to electrical, it is the **only place where drugs can act to affect the nervous system** - eg. this is where heroin works



How an impulse is passed on at a synapse

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